

E1 — Embedded Track

Member: _____

Your role

You own the hardware. You own the board schedule. You bring up the board in week 5 and measure the on-chip numbers.

Weekly tasks

Week 1 — Toolchain and firmware study

- Task: Install the ESP32 toolchain on your machine (Arduino ESP32 core 3.3.10 or ESP-IDF). Build the portable C runtime on host. Read the firmware/ sources.
- Deliverable: The toolchain installed. The firmware walkthrough notes.
- Verify: The host runtime builds without error.
- Learn: ESP32-S3 architecture: 2 cores, 512 KB SRAM, 8 MB PSRAM, 16 MB flash. Draw the memory map.

Week 2 — Deploy pipeline on host

- Task: Run a dry run of the deploy pipeline on host: `fetch_model.sh` and `deploy.sh` without a board. Prepare the week-5 flash plan.
- Deliverable: The flash plan for week 5.
- Verify: The scripts fail cleanly without a board, with a clear message.
- Learn: The deploy pipeline: `model.bin`, headers, compile, flash. Explain the boot flow.

Week 3 — Runtime review

- Task: Review the `src/` runtime for week-5 flash. Write the "what to expect at boot" doc: serial output, timings.
- Deliverable: The boot-expectation doc.
- Verify: H1 reads the doc and finds no missing step.
- Learn: Runtime internals: `int8` head staging, KV cache, the memory budget.

Week 4 — Flash plan finalize

- Task: Finalize the week-5 flash plan with E2. Rehearse the steps on paper.
- Deliverable: The final flash plan.
- Verify: E2 signs off on the plan.
- Learn: Measurement protocol: `tok/s`, `ms/token`, memory free. Write the measurement steps.

Week 5 — Board bring-up

- Task: Flash the stock `tinystories` model: `bash scripts/fetch_model.sh tinystories` and `bash scripts/deploy.sh tinystories`. Verify text generation. Measure `tok/s`, `ms/token`, and memory free.
- Deliverable: The on-chip verification table for the stock model.
- Verify: Your `tok/s` sits within +/-15% of 9.88.
- Learn: On-chip verification. Compare your numbers with the published table.

Week 6 — Our model on chip + hardware gate

- Task: Flash our exported trained model. Measure `tok/s` and memory vs the stock model. Run the hardware gate.
- Deliverable: The comparison table and the gate verdict.

- Verify: The gate criteria pass: tok/s within +/-15%, coherent text.
- Learn: Firmware features: serial prompting, profiling. Read the int8 head code.

Week 7 — Demo hardening

- Task: Harden the demo: power cycles, boot loop, serial reliability. Final board state.
- Deliverable: The hardening report.
- Verify: The board boots 10 times in a row without failure.
- Learn: Demo operations: power, boot, serial. Write the demo checklist.

Week 8 — Live demo

- Task: Run the live demo on hardware at the showcase.
- Deliverable: The live demo.
- Verify: The demo runs during the whole showcase.
- Learn: Write the "what I learned" page.

Learning path

1. ESP32-S3 architecture: cores, SRAM, PSRAM, flash.
2. Deploy pipeline and boot flow.
3. Runtime internals: int8 head, KV cache, memory budget.
4. Measurement protocol: tok/s, ms/token, bandwidth.
5. On-chip verification vs published numbers.
6. Firmware features: serial prompting, profiling.
7. Demo hardening: power, boot, serial reliability.
8. Live demo and final numbers.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- firmware/: <https://github.com/slvDev/esp32-ai/tree/main/firmware>
- RESULTS.md: <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- ESP32-S3 docs: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/>